

Cats: Coefficient of Determination

MATH 213

Group 1

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

1. What is the explanatory (or predictor) variable in my model?

The weight of the cat's heart is the explanatory variable.

1. What is the response variable in my model?

The response variable is the cat's body weight.

```
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

3. From the model output, report the coefficient of determination, R^2 .

4. Write a sentence of interpretation for the R^2 value.

5. What is the value of r , the correlation coefficient?

```
##{r}  
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 2

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?
- 2. What is the response variable in my model?

The cat's body weight is the response variable

```
##{r}
summary(cats_model)

Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:
    Min       1Q   Median       3Q      Max
-3.5694 -0.9634 -0.0921  1.0426  5.1238

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.3567      0.6923  -0.515   0.607
Bwt           4.0341      0.2503  16.119 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared:  0.6466,    Adjusted R-squared:  0.6441
F-statistic: 259.8 on 1 and 142 DF,  p-value: < 2.2e-16
```

3. From the model output, report the coefficient of determination, R^2 .

0.6466 is the coefficient of determination

4. Write a sentence of interpretation for the R^2 value.

About 65% of the variability in heart weight is explained by using body weight as a predictor.

5. What is the value of r , the correlation coefficient?

$\sqrt{0.6466} = 0.80411441971$

```
##{r}
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 3

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

The explanatory/predictor variable in the model is the body weight of each cat.

- 1. What is the response variable in my model?

The response variable in the model is the heart weight of each cat.

```
summary(cats_model)
```



```
Call:
lm(formula = Hwt ~ Bwt, data = cats)
```



```
Residuals:
```

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238


```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***


```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared:  0.6466,    Adjusted R-squared:  0.6441
F-statistic: 259.8 on 1 and 142 DF,  p-value: < 2.2e-16
```

3. From the model output, report the coefficient of determination, R^2 .

The coefficient of determination is 0.6466.

4. Write a sentence of interpretation for the R^2 value.
The body weight of cats explains roughly 65% of the variation in cats' heart weight.

5. What is the value of r , the correlation coefficient?
 $\sqrt{0.6466} \approx 0.8041$.

```
##{r}
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 4

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

Body weight

- 1. What is the response variable in my model?

Heart weight

```
summary(cats_model)
```

```
Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:
    Min       1Q   Median       3Q      Max
-3.5694 -0.9634 -0.0921  1.0426  5.1238

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.3567     0.6923  -0.515   0.607
Bwt           4.0341     0.2503  16.119 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared:  0.6466,    Adjusted R-squared:  0.6441
F-statistic: 259.8 on 1 and 142 DF,  p-value: < 2.2e-16
```


3. From the model output, report the coefficient of determination, R^2 .

0.65

4. Write a sentence of interpretation for the R^2 value.

65% of the variability in heart weight can be explained by the variation in body weight

5. What is the value of r , the correlation coefficient?

0.81

```
##{r}  
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 5

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

The body weight

- 1. What is the response variable in my model?

Weight of the heart

```
summary(cats_model)
```

Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441
F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

3. From the model output, report the coefficient of determination, R^2 .

.6466

4. Write a sentence of interpretation for the R^2 value.

About 65% of the variability in heart weight is explained by the model. The remainder of the variability is explained by variables not included in the model or by inherent randomness in the data.

5. What is the value of r , the correlation coefficient?

.80411441971

```
##{r}  
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 6

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

The predictor variable in the model is the body weight of each cat (kg)

- 1. What is the response variable in my model?

The response variable is the weight of each cats heart (grams).

```
summary(cats_model)
```

Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441
F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

3. From the model output, report the coefficient of determination, R^2 .

$R^2 = 0.6466$

4. Write a sentence of interpretation for the R^2 value.

About 65% of the variability in the weight of the cats heart (grams) is explained by the cats body weight (kg).

5. What is the value of r , the correlation coefficient?

$R = 0.804114419719$

```
##{r}
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 7

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

Body weight(bwt, kg)

- 1. What is the response variable in my model?

Heart weight(hwt, grams)

```
summary(cats_model)
```

Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441
F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

3. From the model output, report the coefficient of determination, R^2 .

0.6466

4. Write a sentence of interpretation for the R^2 value.

The r^2 value indicates that 64.66% of the variability in heart weight is explained by body weight. Additionally with this score being relatively high we can assume there is a strong relationship between body weight and heart weight

5. What is the value of r , the correlation coefficient?

$R = \sqrt{0.6466} = 0.8041$

```
## {r}  
summary(cats_model)
```

Call:

`lm(formula = Hwt ~ Bwt, data = cats)`

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 8

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

1. What is the explanatory (or predictor) variable in my model?

The explanatory variable is body weight

1. What is the response variable in my model?

The response variable is heart weight

```
##{r}  
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

3. From the model output, report the coefficient of determination, R^2 .

0.6466

4. Write a sentence of interpretation for the R^2 value.

About 64.66% of the variability in heart weight is explained by using body weight as a predictor.

5. What is the value of r , the correlation coefficient?

0.804

(r is positive because slope is positive)

```
##{r}
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 9

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

Weight of cat's heart - got these backwards

- 1. What is the response variable in my model?

Body Weight - got these backwards

```
summary(cats_model)

Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:
    Min       1Q   Median       3Q      Max
-3.5694 -0.9634 -0.0921  1.0426  5.1238

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.3567      0.6923  -0.515   0.607
Bwt          4.0341      0.2503  16.119 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared:  0.6466,    Adjusted R-squared:  0.6441
F-statistic: 259.8 on 1 and 142 DF,  p-value: < 2.2e-16
```

3. From the model output, report the coefficient of determination, R^2 .

.6466

4. Write a sentence of interpretation for the R^2 value.

About 64% of the variability in body weight is explained by using the weight of the cat's heart as a predictor.

5. What is the value of r , the correlation coefficient?

.8041, positive for a positive slope

```
##{r}  
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Group 10

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (Hwt), in grams, the body weight (Bwt) of each cat, in kg, and the sex of each cat, (Sex).

To the right is some summary output from a least squares regression model I made.

- 1. What is the explanatory (or predictor) variable in my model?

The body weight is the predictor

- 1. What is the response variable in my model?

The weight of their heart

```
summary(cats_model)
```

```
Call:
lm(formula = Hwt ~ Bwt, data = cats)

Residuals:
    Min       1Q   Median       3Q      Max
-3.5694 -0.9634 -0.0921  1.0426  5.1238

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.3567     0.6923  -0.515   0.607
Bwt           4.0341     0.2503  16.119 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom
Multiple R-squared:  0.6466,    Adjusted R-squared:  0.6441
F-statistic: 259.8 on 1 and 142 DF,  p-value: < 2.2e-16
```

3. From the model output, report the coefficient of determination, R^2 .

.65

4. Write a sentence of interpretation for the R^2 value.

About .65 of the variability of the heart weight is explained by using total body weight

5. What is the value of r , the correlation coefficient?

.806

```
##{r}  
summary(cats_model)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16